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Roll Number : 31/A		Lab Practical Number: 08	
Title of LAB Practical : Implementation of Storage as a Service using Google Docs			
DOP : 25/09/2025		DOS : 12/10/2025	
CO Mapped : CO5, CO6, CO7	PO Mapped: PO1, PO2, PO4, PO7, PSO1, PSO2	Faculty Signature:	Marks :

Title: Implementation of SaaS using Google Docs

Aim: To implement Storage as a Service using Google Docs

Theory:

Software as a Service (SaaS) is a cloud computing model where software applications are hosted by a service provider and made available to users over the internet. Instead of installing and maintaining software locally, users can access it through a web browser. Google Docs is a leading example of SaaS in action.

1. Working Principle of Google Docs as SaaS:

- Web-based Application:

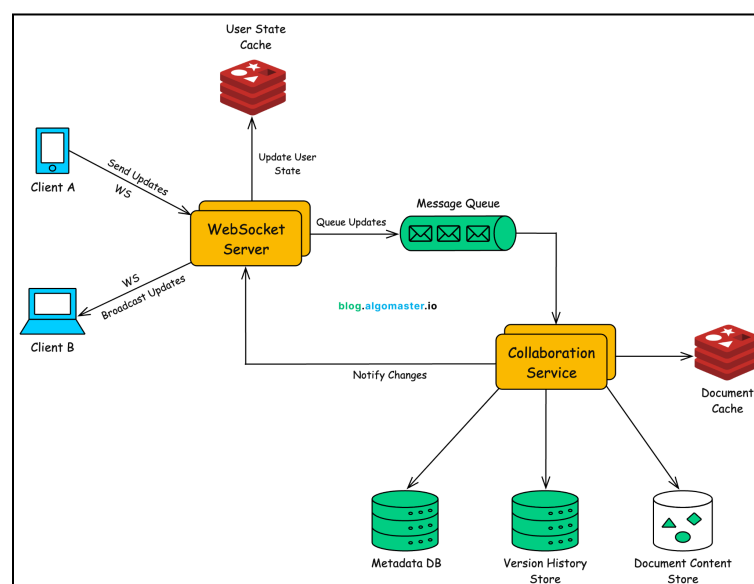
Google Docs runs entirely within a web browser. The user interface and processing logic are delivered from Google's cloud servers to the user's device.

- Centralized Hosting:

The application is centrally hosted on Google Cloud. All updates, maintenance, and bug fixes are handled by Google, not the end user.

- Internet-based Access:

Users can access their documents from any device connected to the internet using their Google account.



This diagram explains how real-time collaboration works in a SaaS application like Google Docs — allowing multiple users to edit a document at the same time.

1. Clients (Client A and Client B):

- These are users working on the same document (e.g., two people editing a Google Doc).
- Each client connects to the server using a WebSocket (WS) connection for real-time updates.

2. WebSocket Server:

- It acts as a communication hub between all connected clients.
- When Client A makes a change, it:
 - Sends the update to the WebSocket Server.
 - The server queues the update and sends it to the Collaboration Service.
- The server then broadcasts updates to other connected clients (like Client B), ensuring everyone sees the latest version.

3. Message Queue:

- Stores and manages updates temporarily before they are processed.
- Ensures that document changes are handled in the correct order — even if multiple users send edits simultaneously.

4. Collaboration Service:

- The brain of the system — responsible for merging, validating, and synchronizing all document edits.
- It takes updates from the message queue and applies them to the document.
- Then it notifies the WebSocket Server of changes to broadcast to all clients.

5. Caches and Databases:

- User State Cache: Keeps track of which user is online, editing, or idle.
- Document Cache: Stores the latest version of the document for quick access.
- Metadata DB: Stores information about the document (like title, authors, timestamps).
- Version History Store: Keeps a record of all past versions and changes for undo/redo or recovery. Document Content Store: Contains the actual text and structure of the document.

6. Workflow Summary:

1. Client A edits a document → sends update via WebSocket.
2. WebSocket Server sends update to Collaboration Service through the message queue.
3. Collaboration Service processes and saves the update to storage.
4. Collaboration Service notifies WebSocket Server.
5. WebSocket Server broadcasts the change to all connected clients (like Client B).
6. All users see the same updated document instantly.

2. Key Components of Implementation:**A. Cloud Infrastructure:**

Google uses its own Google Cloud Platform (GCP) to host and manage Google Docs. This includes compute instances, databases, load balancers, and distributed storage systems.

B. Multi-tenancy Model:

All users share the same software infrastructure but have isolated and secure document storage spaces. This allows efficient resource utilization and scalability.

C. Data Storage and Synchronization:

Documents created in Google Docs are stored in Google Drive.

Every edit is automatically synchronized and saved in real-time on Google's servers. This eliminates the risk of data loss.

D. Real-time Collaboration Engine:

Google Docs implements collaborative editing using Operational Transformation (OT) and real-time database synchronization. This allows multiple users to edit the same document simultaneously and view changes instantly.

E. Security and Privacy:

Data is encrypted using HTTPS during transmission and AES encryption while stored. Access control and authentication are managed through Google Account credentials and permissions (view, comment, edit).

4. Features Enabled by SaaS Model:

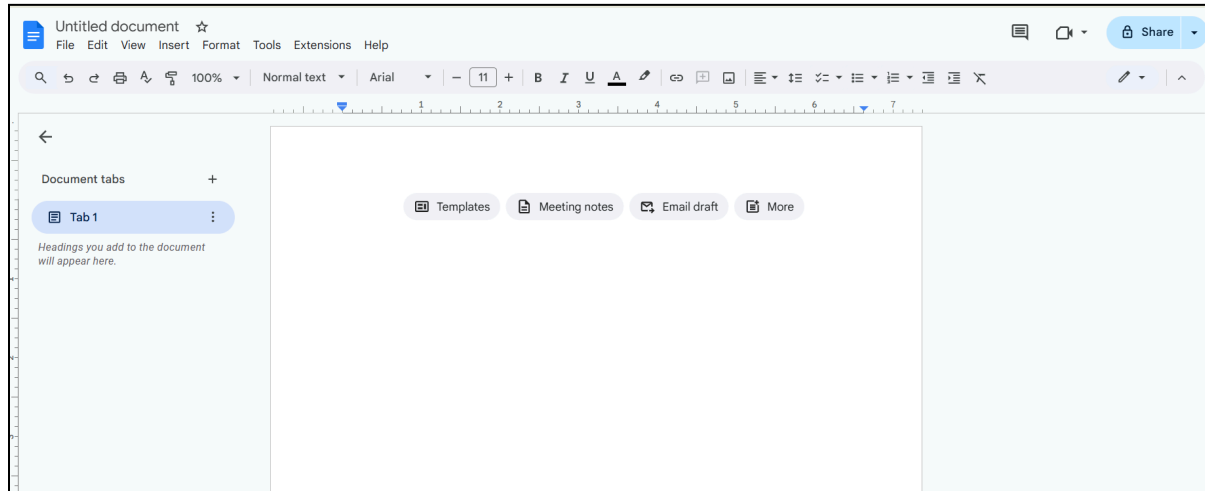
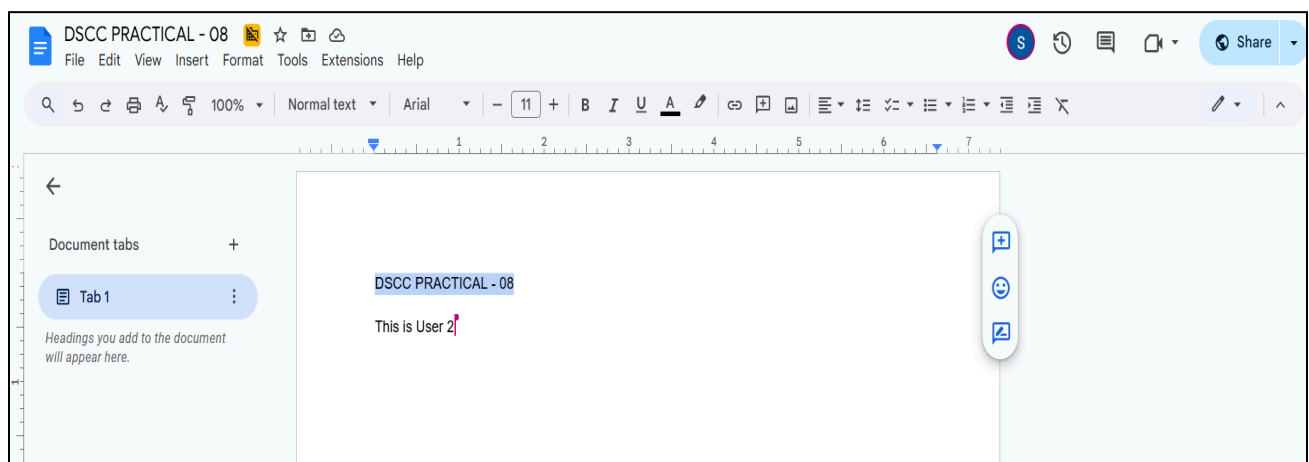
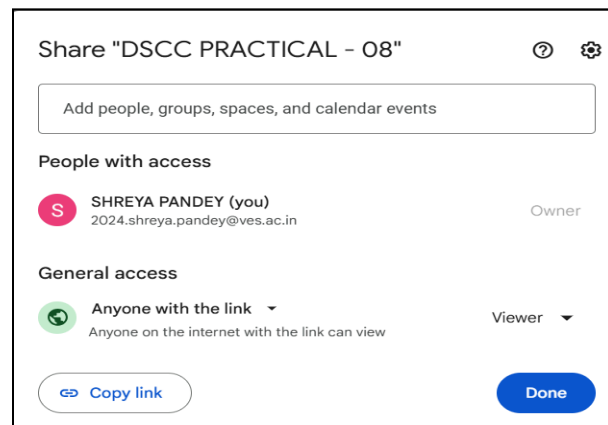
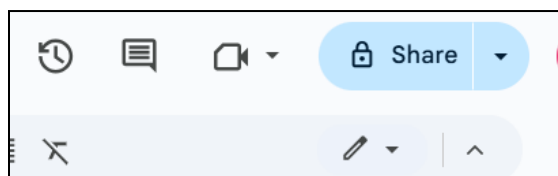
- Scalability: Easily supports millions of concurrent users without local setup.
- Accessibility: Works on all platforms — Windows, macOS, Android, iOS — through browsers or apps.
- Automatic Updates: New features and security patches are added without user action.
- Collaboration Tools: Built-in chat, comments, and sharing permissions for teamwork.
- Offline Mode: Allows document editing without internet; changes sync once online.

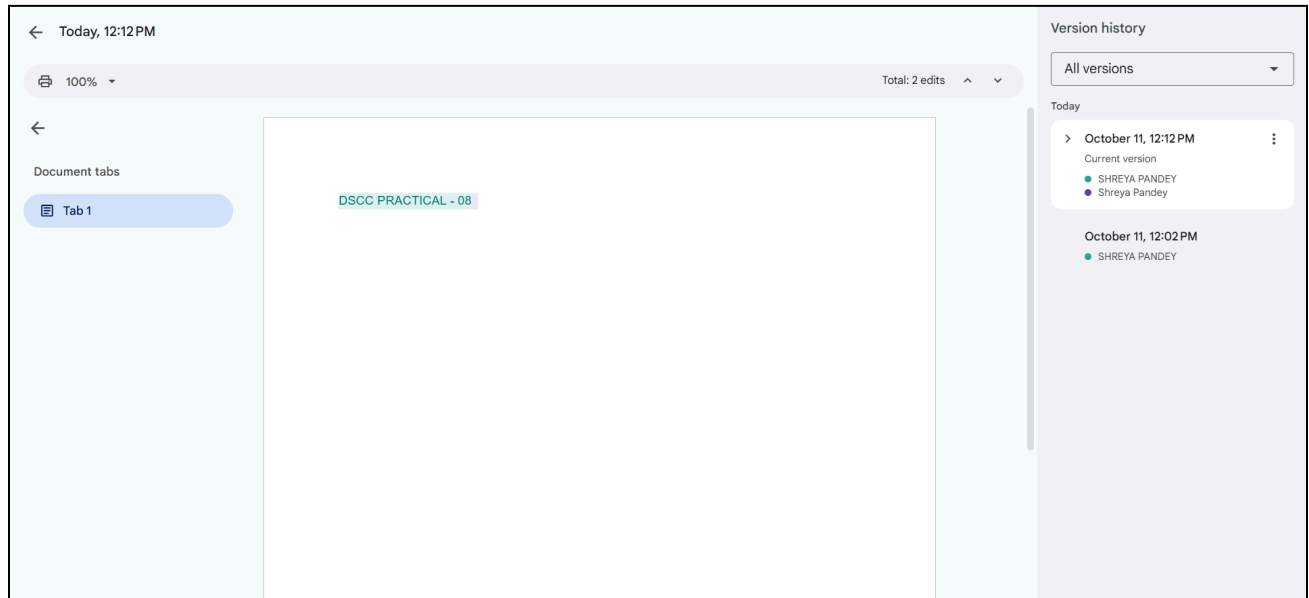
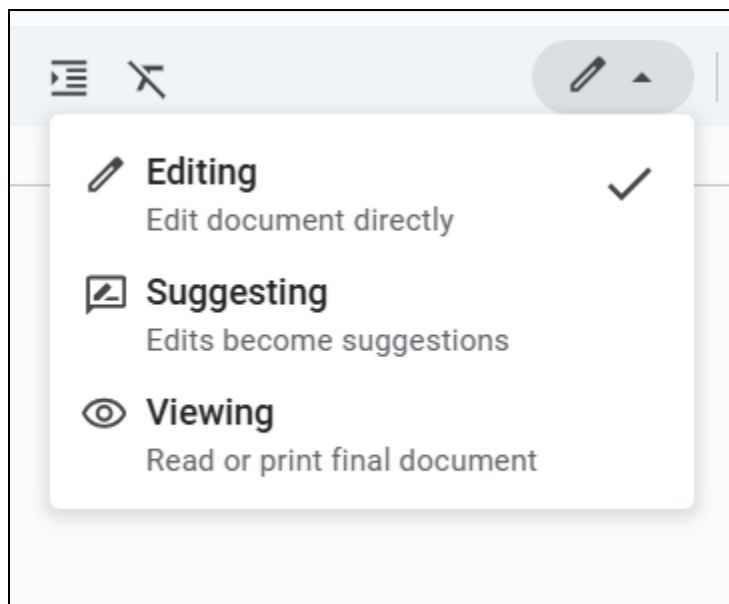
5. Benefits of SaaS in Google Docs

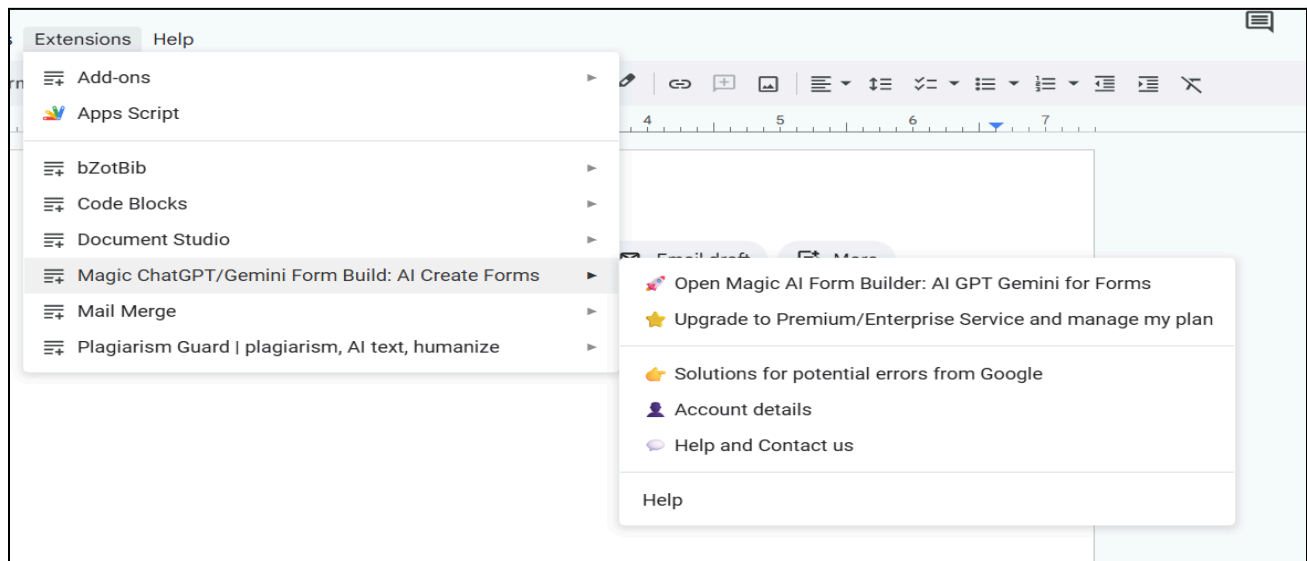
- Reduces cost (no installation or license fee).
- Ensures high availability and uptime.
- Simplifies maintenance and upgrades.
- Promotes team collaboration and productivity.
- Provides data redundancy and backup through cloud storage.

6. Example Flow:

1. User logs in to Google Docs using a Google account.
2. Application loads from Google servers into the browser.
3. User creates or opens a document stored on Google Drive.
4. Edits are instantly synced to the cloud and visible to collaborators.
5. Data remains stored securely on Google's servers for future access.

Google Docs Workspace:**Share the Document so that others can collaborate:**

Versions of Latest commits are available:**Different Modes available:**

Google Docs Extensions:

Magic AI Form Builder:
GPT/Gemini for Forms

Select what to create

☒ Convert Doc/Slides/Sheets to Form

☐ Create form with provided text/topic

☐ Create form from file (Upgrade and use)

☐ Create form from website url

☐ Generate multiple choices for your questions

☐ Import your questions in bulk (Needs upgrade)

1. Select what to convert to form and add optional needs

☒ Select the text

☐ Entire Doc (Needs upgrade)

Enter the name of the form to create

DSCC

2. What kind of questions do you want?

Question Type:

Multiple Choice

2. What kind of questions do you want?

Question Type:

Multiple Choice

How many questions: 2

How many choices per question:

4

How many answers each question (optional):

1

How difficult (optional)

College F

☐ Make as Quiz (with points)

Create Form

Created Form: new_form_from_Untitled document

1. What is the primary focus of the DSCC PRACTICAL - 08 course?
a) Analyzing theoretical concepts
b) Understanding practical applications of digital systems
c) Developing software programming skills
d) Conducting laboratory experiments
Answer: Understanding practical application of digital systems

2. Which of the following best describes the expected outcome of completing DSCC PRACTICAL - 08?
a) Ability to apply theoretical knowledge in real-world scenarios
b) Mastery of advanced programming languages
c) Development of hardware engineering skill
d) Expertise in data analysis techniques
Answer: Ability to apply theoretical knowledge in real-world scenarios

Conclusion:

The implementation of SaaS using Google Docs demonstrates how cloud technology revolutionizes software delivery. By shifting from traditional installations to web-based access, Google Docs provides a scalable, collaborative, and secure platform that exemplifies the true power and efficiency of Software as a Service.